

MAHMUD HASAN MUNNA

Senior Data Scientist | AI/ML Engineer

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PROFESSIONAL SUMMARY

Senior Data Scientist and ML Engineer with 4+ years of experience architecting and deploying production-grade machine learning systems at scale. Proven track record in end-to-end ML lifecycle management—from data pipeline engineering and feature design to model training, deployment, monitoring, and automated retraining. Hands-on expertise in MLOps, CI/CD for ML, LLM/GenAI systems (RAG, Agentic AI), time-series forecasting, geospatial analytics, and NLP. Telecom-domain experience processing 1B+ records. Skilled in Python, TensorFlow, PyTorch, Airflow, Docker, FastAPI, and AWS.

PROFESSIONAL EXPERIENCE

Wegro Technologies Limited — Senior Data Scientist

Aug 2024 – Present

- Architected a company-wide centralized data warehouse and automated ETL pipelines using PostgreSQL, Pandas, Airflow, AWS EC2/RDS/S3, and Docker with full CI/CD — enabling reliable batch and near-real-time data flows.
- Designed and deployed an eKYC onboarding module using Computer Vision, OCR, and LLMs (FastAPI + AWS), reducing farmer onboarding time by 90% and eliminating fake registrations.
- Built a Business Intelligence Agentic AI Chatbot (LangChain, LangGraph, WebSocket API, PostgreSQL) enabling natural-language database querying, cutting surface-level analytics turnaround by ~70%.
- Automated 30+ recurring financial and operational reports via ETL pipelines and Plotly-Dash dashboards, saving ~80 hours/month of manual work.
- Delivered production-ready credit scoring models and analytics dashboards, improving business decision-making efficiency by ~50%.
- Built a Terms & Conditions compliance validator using RAG, LangChain, and FAISS vector database — deployed on AWS with FastAPI, Docker, and GitHub Actions CI/CD.
- Implemented an automated Meeting Minutes Summarizer using Whisper (speech-to-text), DialoGPT (speaker separation), and LLM summarization.

SSL Wireless — Data Scientist

Mar 2022 – Aug 2024

- Engineered ETL pipelines processing 1B+ customer behavior records using Dask and Apache Airflow, generating targeted marketing and credit scoring insights for large payment gateway clients.
- Built and deployed an NLP-based SMS Compliance Validator achieving 95% accuracy, handling 100,000+ messages/hour in production via FastAPI and Docker.
- Developed a geospatial fraud detection system adopted by 300+ field agents, significantly improving field audit efficiency using geospatial ML models, FastAPI, and CI/CD pipelines.
- Designed and deployed telecom recharge forecasting and analytics pipelines using Neural Prophet and time-series models (FastAPI, Docker, CI/CD), reducing balance-out incidents for telecom operations.
- Delivered end-to-end ML system for mitigating checkout abandonment in e-commerce (published at FedCSIS 2022).

CORE COMPETENCIES & TECHNICAL SKILLS

ML & AI	Supervised/Unsupervised Learning, Ensemble Methods, DNN/CNN/RNN, Time-Series Forecasting, Anomaly Detection, Recommendation Systems, NLP, Computer Vision
GenAI / LLMs	LangChain, LangGraph, RAG, Vector Databases (FAISS), Prompt Engineering, Agentic AI, Whisper, Transformers
MLOps & Deploy	MLflow (Experiment Tracking), Docker, CI/CD (GitHub Actions), FastAPI, OAuth2, JWT, Model Monitoring, Automated Retraining
Data & ETL	Apache Airflow, Dask, Pandas, Polars, NumPy, MySQL, PostgreSQL, AWS (EC2, RDS, S3), Batch & Streaming Pipelines
ML Frameworks	TensorFlow, PyTorch, Scikit-learn, Neural Prophet, OpenCV, OCR
Visualization	Plotly-Dash, Seaborn, Streamlit, Looker Studio
Languages	Python (primary), SQL, C, MATLAB
Cloud	AWS EC2, RDS, S3; familiarity with cloud-native ML deployment and microservices architecture

KEY ML PROJECTS

Geospatial Fraud Detection for FMCG Supply Chain | Skills: Geospatial Analysis, Traditional ML, FastAPI, Docker, CI/CD
Telecom Recharge Forecasting (Easy.com) | Skills: Time-Series, Neural Prophet, FastAPI, Docker, CI/CD
Billion-Scale Customer Behavior ETL & Credit Scoring | Skills: ETL, MySQL, Dask, Airflow, DNN, TensorFlow, Docker
Agentic BI Chatbot (DB Query via Human Prompt) | Skills: LangChain, LangGraph, PostgreSQL, WebSocket API, FastAPI
eKYC Farmer Onboarding (Computer Vision + LLM) | Skills: OCR, LLM, AWS EC2/RDS/S3, FastAPI, Docker, CI/CD
RAG-based Terms & Conditions Validator | Skills: LangChain, FAISS, AWS, FastAPI, Docker

EDUCATION

B.Sc. in Electronics & Telecommunication Engineering

2016 – 2021

Rajshahi University of Engineering and Technology (RUET), Rajshahi

PEER-REVIEWED PUBLICATIONS

- M. H. Munna et al., "Sentiment Analysis and Product Review Classification in E-commerce," ICCIT 2020, doi: 10.1109/ICCIT51783.2020.9392710
- M. R. I. Rifat, M. H. Munna et al., "End-to-end ML System for Mitigating Checkout Abandonment," FedCSIS 2022, doi: 10.15439/2022F167
- M. H. Munna & M. S. Hossen, "Identification of Clickbait in Video Sharing Platforms," ACMI 2021, doi: 10.1109/ACMI53878.2021.9528095

ACHIEVEMENTS & CERTIFICATIONS

- Grand Finalist — Robi Datathon 3.0, organized by Robi Axiata Ltd. (2024)
- Training on Artificial Intelligence — PUM Netherlands Senior Experts / BASIS (2023)
- Cloud Journey with AWS — BASIS (2023)